

SOPRON, Hungary

WMO: 128050

Lat: 47.6833N

Lon: 16.6000E

Elev: 239

StdP: 98.49

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.8	-7.5	-14.2	1.1	-8.2	-11.4	1.5	-5.5	12.3	4.5	11.1	3.1	3.2	310	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.0	21.5	30.1	20.9	28.4	20.0	22.6	29.6	21.7	28.5	20.8	27.1	4.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.3	15.4	26.2	19.3	14.5	25.2	18.4	13.7	24.2	67.8	29.5	64.2	28.6	61.0	27.2	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.9	7.9	-12.7	34.4	3.3	2.1	-15.1	36.0	-17.1	37.2	-18.9	38.4	-21.3	40.0		
(5)			-13.3	24.2	3.2	1.0	-15.7	24.9	-17.6	25.5	-19.4	26.1	-21.8	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	0.2	2.3	6.3	11.4	15.8	19.5	21.3	20.8	15.9	10.9	5.8	1.0		
	DBStd	8.35	4.36	4.55	4.34	4.00	3.47	3.45	3.10	3.13	3.29	3.84	3.98	4.05		
	HDD10.0	1130	303	217	128	31	2	0	0	0	1	36	134	279		
	HDD18.3	2963	561	449	373	208	94	28	9	12	87	230	376	537		
	CDD10.0	1491	1	2	13	74	180	284	351	334	178	64	9	1		
	CDD18.3	282	0	0	0	1	14	62	102	88	14	1	0	0		
	CDH23.3	2494	0	0	0	22	151	552	901	752	112	6	0	0		
	CDH26.7	783	0	0	0	2	23	164	316	260	19	0	0	0		
(14) Wind	WSAvg	3.3	3.5	3.8	3.9	3.8	3.5	3.3	3.1	2.8	3.0	2.9	3.2	3.3		
(15) Precipitation	PrecAvg	666	29	28	41	45	73	86	79	76	67	54	52	37		
	PrecMax	839	64	75	90	89	176	191	198	160	170	122	112	78		
	PrecMin	501	4	2	3	1	19	25	1	12	12	5	4	3		
	PrecStd	97	16	21	23	22	37	40	46	38	39	30	26	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	16.2	20.8	26.0	29.0	32.9	34.1	34.1	29.1	24.0	18.4	13.5		
		MCWB	8.4	9.5	11.8	16.2	18.7	21.8	22.0	21.9	19.7	17.1	13.7	9.6		
	2%	DB	10.4	13.1	18.1	23.2	26.9	30.5	32.0	31.9	26.2	21.2	15.6	10.3		
		MCWB	6.7	8.2	10.8	14.5	18.0	21.4	21.6	21.3	19.0	15.9	12.3	7.5		
	5%	DB	8.2	11.1	15.5	21.0	25.0	28.5	30.1	29.7	24.3	19.1	13.3	8.2		
		MCWB	5.2	7.2	9.6	13.3	17.3	20.2	20.9	20.8	17.8	14.8	10.8	5.6		
	10%	DB	6.4	8.9	13.2	18.9	22.9	26.7	28.2	27.6	22.2	16.8	11.4	6.6		
		MCWB	4.2	5.7	8.4	12.2	16.0	19.2	20.1	19.9	16.7	13.6	9.3	4.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	10.0	12.7	16.7	20.4	23.8	23.7	23.8	20.9	17.9	14.5	9.9		
		MCDB	12.2	15.2	18.9	24.6	26.8	30.1	31.1	30.9	27.2	22.6	17.4	12.7		
	2%	WB	7.1	8.6	11.3	15.2	18.9	22.0	22.6	22.5	19.6	16.5	12.5	7.7		
		MCDB	9.8	12.6	16.9	21.9	25.2	28.7	29.7	29.6	25.1	20.3	15.1	9.9		
	5%	WB	5.6	7.3	10.1	13.9	17.8	20.9	21.6	21.5	18.5	15.3	11.1	6.1		
		MCDB	7.6	10.7	14.8	19.7	23.7	27.1	28.5	28.2	23.0	18.2	13.1	7.8		
	10%	WB	4.4	5.9	8.8	12.7	16.7	19.8	20.8	20.4	17.4	14.0	9.6	4.9		
		MCDB	6.3	8.6	12.6	18.0	21.9	25.7	27.2	26.4	21.3	16.5	11.1	6.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	7.3	9.0	10.5	10.5	10.6	10.8	10.8	9.6	8.8	6.2	5.5		
		MCDBR	8.9	10.9	13.3	14.4	13.7	13.5	14.2	14.3	12.9	11.9	9.6	8.9		
	5% WB	MCWBR	6.2	6.9	7.5	7.3	6.8	6.1	6.2	6.2	6.6	7.1	6.5	6.4		
		MCWBR	8.3	10.3	12.4	12.9	12.6	12.8	12.9	13.2	11.9	10.9	9.1	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.338	0.376	0.419	0.417	0.421	0.427	0.428	0.402	0.373	0.345	0.314		
		taud	2.398	2.361	2.285	2.188	2.242	2.264	2.249	2.260	2.305	2.362	2.396	2.424		
	Edn at Noon		760	817	837	832	848	845	833	814	799	763	711	713		
		Edh at Noon	69	89	111	135	133	131	131	125	109	88	70	61		
(44) All-Sky Solar Radiation	RadAvg		1.05	1.86	3.07	4.53	5.37	5.84	5.70	5.06	3.62	2.26	1.19	0.81		
	RadStd		0.11	0.26	0.38	0.44	0.55	0.41	0.47	0.49	0.42	0.23	0.11	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (82 neighbors)		+0.57	N/A	N/A	+0.88	+0.48	+0.41	-117	-160	+92	+49

Nomenclature: See separate page